

Score Matching

Assume data x follows some unknown distribution $x \sim p(x)$. The **score function** is the derivative of the log density:

$$s(x) := \nabla_x \log p(x).$$

Why score function

The most natural target for a generative model to learn is the probability density function of the data $p(x)$. Suppose $p_\theta(x)$ is a neural network to approximate $p(x)$. One way (energy-based model) to design this network is to let $f_\theta(x)$ be a real-valued function (a neural network) and put

$$p_\theta(x) = \frac{e^{-f_\theta(x)}}{Z_\theta},$$

where Z_θ is the normalizing constant so that $\int p_\theta(x) dx = 1$. We train the network by maximizing the log-likelihood:

$$\theta^* = \arg \max_{\theta} \sum_{i=1}^N \log p_\theta(x_i).$$

This plan has a problem: the normalizing constant Z_θ is in general intractable. The score function $s(x) = \nabla_x \log p(x)$ does not have this issue: if we put $p_\theta(x) = \frac{e^{-f_\theta(x)}}{Z_\theta}$, then $s_\theta(x) = -\nabla_x f_\theta(x)$ which does not involve Z_θ .

“This significantly expands the family of models that we can tractably use, since we don’t need any special architectures to make the normalizing constant tractable.”

How score function generates

Once we obtained the score function $s(x)$, **Langevine dynamics** allows us to draw samples from $p(x)$. It is an iterative process: let $x_0 \sim \pi(x)$ be drawn from an arbitrary prior distribution, and then it iterates through

$$x_{i+1} = x_i + \alpha s(x_i) + \eta z_i,$$

where $\alpha, \eta > 0$ and $z_i \sim \mathcal{N}(0, I)$. It converges to some $\tilde{x}$ with $p(\tilde{x}) > 0$ and therefore a sample from $p(x)$.

How to learn score function

Let $s_\theta(x)$ be a neural network approximating the score function $s(x)$. We try to minimize the Fisher divergence:

$$J(\theta) = \frac{1}{2} \mathbb{E}_{p(x)} \|s_\theta(x) - \nabla_x \log p(x)\|^2.$$

Note that the gradient $\nabla_x \log p(x)$ is exactly what we need to learn. To make the objective tractable, we have to find a way around this.

Score Matching

We note:

$$\begin{aligned} J(\theta) &= \frac{1}{2} \mathbb{E}_{p(x)} \|s_\theta(x) - \nabla_x \log p(x)\|^2 \\ &= \frac{1}{2} \mathbb{E}_{p(x)} [\|s_\theta(x)\|^2 - 2s_\theta(x)^\top \nabla_x \log p(x) + \|\nabla_x \log p(x)\|^2]. \end{aligned}$$

The last term does not involve θ and therefore is ignored.

The middle term is

$$\begin{aligned}
& \mathbb{E}_{p(x)} [s_\theta(x)^\top \nabla_x \log p(x)] \\
&= \int s_\theta(x)^\top (\nabla_x \log p(x)) p(x) \, dx \\
&= \int s_\theta(x)^\top \left(\frac{\nabla_x p(x)}{p(x)} \right) p(x) \, dx \\
&= \int s_\theta(x)^\top \nabla_x p(x) \, dx.
\end{aligned}$$

The integral, expressed coordinate-wise, is

$$\begin{aligned}
& \int \sum_i s_\theta(x)_i \cdot \frac{\partial p(x)}{\partial x_i} \, dx \\
&= \sum_i \int s_\theta(x)_i \cdot \frac{\partial p(x)}{\partial x_i} \, dx.
\end{aligned}$$

It can be shown, by an argument similar to integration by parts, that

$$\int s_\theta(x)_i \cdot \frac{\partial p(x)}{\partial x_i} \, dx = - \int \frac{\partial s_\theta(x)_i}{\partial x_i} \cdot p(x) \, dx$$

under “weak regularity conditions”. (The conditions are, for all θ , $\mathbb{E}_{p(x)} \|s_\theta(x)\|^2 < \infty$ and $s_\theta(x)p(x) \rightarrow 0$ when $\|x\| \rightarrow \infty$.)

Hence,

$$\begin{aligned}
& \mathbb{E}_{p(x)} [s_\theta(x)^\top \nabla_x \log p(x)] \\
&= - \int \sum_i \frac{\partial s_\theta(x)_i}{\partial x_i} \cdot p(x) \, dx \\
&= - \mathbb{E}_{p(x)} [\text{Tr}(\nabla_x s_\theta(x))].
\end{aligned}$$

Putting together, up to a constant (w.r.t θ), the objective becomes

$$J(\theta) = \mathbb{E}_{p(x)} \left[\frac{1}{2} \|s_\theta(x)\|^2 + \text{Tr}(\nabla_x s_\theta(x)) \right].$$

In this way we get rid of the gradient $\nabla_x \log p(x)$ in the original expression of $J(\theta)$.

This method, however, still has two issues:

(1) computing the trace $\text{Tr}(\nabla_x s_\theta(x))$ is costly. The Hutchinson's estimator

$$\text{Tr}(A) = \mathbb{E}_\nu [\nu^\top A \nu], \quad \nu \sim \mathcal{N}(0, I)$$

may help.

(2) this estimation of gradients is inaccurate in low-density areas. The real data may have very limited support. The estimation is inconsistent off the support.

Denoising Score Matching

We perturb data x by $\tilde{x} = x + \sigma z$ for $\sigma > 0$ and $z \sim \mathcal{N}(0, I)$ independent from x . The density function of $\tilde{x}$, $p_\sigma(\tilde{x})$, is the convolution

$$p_\sigma(\tilde{x}) = [p * \mathcal{N}(0, \sigma I)](\tilde{x}) = \int p(x) \mathcal{N}(\tilde{x}; x, \sigma I) dx.$$

The objective function is

$$J(\theta) = \frac{1}{2} \mathbb{E}_{p_\sigma(\tilde{x})} \|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\|^2.$$

We claim

$$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) = \mathbb{E}_{p(x|\tilde{x})} \left[-\frac{\tilde{x} - x}{\sigma^2} \right]. \quad (*)$$

To see (*), we note that

$$\begin{aligned} \nabla_{\tilde{x}} p_\sigma(\tilde{x}) &= \nabla_{\tilde{x}} \int p(x) \mathcal{N}(\tilde{x}; x, \sigma I) dx \\ &= \int p(x) \nabla_{\tilde{x}} \mathcal{N}(\tilde{x}; x, \sigma I) dx. \end{aligned}$$

Moreover,

$$\begin{aligned}\nabla_{\tilde{x}} \mathcal{N}(\tilde{x}; x, \sigma I) &= \mathcal{N}(\tilde{x}; x, \sigma I) \nabla_{\tilde{x}} \log \mathcal{N}(\tilde{x}; x, \sigma I) \\ &= \mathcal{N}(\tilde{x}; x, \sigma I) \left(-\frac{\tilde{x} - x}{\sigma^2} \right).\end{aligned}$$

Hence,

$$\begin{aligned}\nabla_{\tilde{x}} \log p_{\sigma}(\tilde{x}) &= \frac{\nabla_{\tilde{x}} p_{\sigma}(\tilde{x})}{p_{\sigma}(\tilde{x})} \\ &= \int \left(-\frac{\tilde{x} - x}{\sigma^2} \right) \frac{p(x) \mathcal{N}(\tilde{x}; x, \sigma I)}{p_{\sigma}(\tilde{x})} dx \\ &= \int \left(-\frac{\tilde{x} - x}{\sigma^2} \right) p(x|\tilde{x}) dx \\ &= \mathbb{E}_{p(x|\tilde{x})} \left[-\frac{\tilde{x} - x}{\sigma^2} \right],\end{aligned}$$

proving (*).

(*) is a remarkable identity: “the score of the noisy marginal is the posterior expectation of the denoising direction. The local gradient of the log-density at a noisy point $\tilde{x}$ is a weighted average of the directions pointing back toward each plausible clean data point x , with weights given by the posterior $p(x|\tilde{x})$. The score thus encodes where the data is, averaged over posterior uncertainty about which clean point generated $\tilde{x}$. Learning the score is equivalent to learning how to denoise.”

We therefore update the objective function

$$\begin{aligned}J_{\text{DSM}}(\theta) &= \frac{1}{2} \mathbb{E}_{p_{\sigma}(\tilde{x})} \mathbb{E}_{p(x|\tilde{x})} \left\| s_{\theta}(\tilde{x}) + \frac{\tilde{x} - x}{\sigma^2} \right\|^2 \\ &= \frac{1}{2} \mathbb{E}_{x \sim p(x)} \mathbb{E}_{z \sim \mathcal{N}(0, I)} \left\| s_{\theta}(x + \sigma z) + \frac{z}{\sigma} \right\|^2.\end{aligned}$$

$J_{\text{DSM}}(\theta)$ and $J(\theta)$ only differ by a constant w.r.t θ because of (*).

Connection to diffusion models

Instead of perturbing the data with some fixed noise level σ , we may have a time-dependent noise level $\sigma(t)$. In diffusion models, at time t , the perturbed data is

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}z$$

for noise schedule α_t . The objective J_{DSM} is equivalent to

$$L(\theta) = \mathbb{E}_{\substack{x \sim p(x) \\ z \sim \mathcal{N}(0, I) \\ t \sim U[0, T]}} ||\varepsilon_\theta(x_t, t) - z||^2$$

by setting $s_\theta = -\varepsilon_\theta/\sqrt{1 - \bar{\alpha}_t}$, which is the objective used in the diffusion model.